

5.3 Topic 1: Lifestyle, health and risk

Students will be assessed on their ability to:

- 1 Demonstrate knowledge and understanding of the practical and investigative skills identified in numbers 4 and 5 in the table of *How Science Works* on page 12 of this specification.
- 6 Explain why many animals have a heart and circulation (mass transport to overcome limitations of diffusion in meeting the requirements of organisms).
- 2 Explain the importance of water as a solvent in transport, including its dipole nature.
- 8 Explain how the structures of blood vessels (capillaries, arteries and veins) relate to their functions.
- 7 Describe the cardiac cycle (atrial systole, ventricular systole and diastole) and relate the structure and operation of the mammalian heart to its function, including the major blood vessels.
- 11 Explain the course of events that leads to atherosclerosis (endothelial damage, inflammatory response, plaque formation, raised blood pressure).
- 10 Describe the blood clotting process (thromboplastin release, conversion of prothrombin to thrombin and fibrinogen to fibrin) and its role in cardiovascular disease (CVD).
- 12 Describe the factors that increase the risk of CVD (genetic, diet, age, gender, high blood pressure, smoking and inactivity).
- 18 Analyse and interpret quantitative data on illness and mortality rates to determine health risks (including distinguishing between correlation and causation and recognising conflicting evidence).
- 19 Evaluate design of studies used to determine health risk factors (including sample selection and sample size used to collect data that is both valid and reliable).
- 20 Explain why people's perceptions of risks are often different from the actual risks (including underestimating and overestimating the risks due to diet and other lifestyle factors in the development of heart disease).

- 17 Analyse data on energy budgets and diet so as to be able to discuss the consequences of energy imbalance, including weight loss, weight gain, and development of obesity.
- 3 Distinguish between monosaccharides, disaccharides and polysaccharides (glycogen and starch – amylose and amylopectin) and relate their structures to their roles in providing and storing energy (β -glucose and cellulose are not required in this topic).
- 4 Describe how monosaccharides join to form disaccharides (sucrose, lactose and maltose) and polysaccharides (glycogen and amylose) through condensation reactions forming glycosidic bonds, and how these can be split through hydrolysis reactions.
- 5 Describe the synthesis of a triglyceride by the formation of ester bonds during condensation reactions between glycerol and three fatty acids and recognise differences between saturated and unsaturated lipids.
- 14 Analyse and interpret data on the possible significance for health of blood cholesterol levels and levels of high-density lipoproteins (HDLs) and low-density lipoproteins (LDLs). Describe the evidence for a causal relationship between blood cholesterol levels (total cholesterol and LDL cholesterol) and CVD.
- 9 **Describe how the effect of caffeine on heart rate in *Daphnia* can be investigated practically, and discuss whether there are ethical issues in the use of invertebrates.**
- 16 **Describe how to investigate the vitamin C content of food and drink.**
- 15 Discuss how people use scientific knowledge about the effects of diet (including obesity indicators), exercise and smoking to reduce their risk of coronary heart disease.
- 13 Describe the benefits and risks of treatments for CVD (antihypertensives, plant statins, anticoagulants and platelet inhibitory drugs).